

Desain dan Implementasi Sistem Pengamanan Komunikasi Suara Berbasis Enkripsi Frekuensi pada FPGA

Implementasi Sekuens DTMF Satu Arah: Simulasi Integrasi Menyeluruh, Sintesis *Bitstream*, dan
Hardware Loopback Test
29 April - 6 Mei 2026

Tujuan

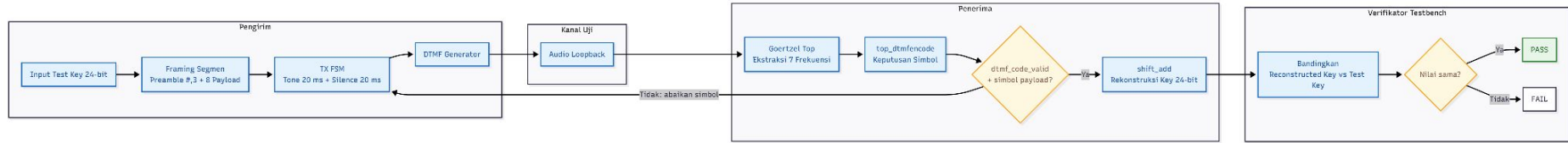
1. Melakukan **simulasi** komunikasi *end-to-end* transmisi kunci 24-bit via sekuens DTMF berbasis **testbench** integrasi *top-level* menggunakan **ModelSim**.
2. Melakukan **sintesis** RTL, **kompilasi** penuh, dan pembuatan *file bitstream* menggunakan Quartus untuk persiapan uji perangkat keras.
3. Mengunggah fail **bitstream** hasil sintesis ke papan **FPGA** Terasic DE10-Standard.
4. Melakukan **Hardware Loopback Test** secara lokal untuk memvalidasi bahwa integrasi subsistem enkripsi dan dekripsi suara dapat beroperasi secara *real-time*.

Aktivitas

Simulasi *Testbench*

- Membuat ***testbench*** integrasi global untuk menguji alur lengkap sistem, mulai dari **pengiriman** simbol DTMF sampai kunci 24-bit berhasil **direkonstruksi** kembali dengan target utama *key* keluaran harus **sama** dengan *key* masukan.
- Mengatur **stimulus** uji (*reset*, input *key*, *start* transmisi, durasi *tone-silence*) dan menyinkronkan dengan *sample tick* agar *timing* pengirim-penerima **sesuai** dan hasil uji **stabil**.
- Menghubungkan seluruh blok **penerima** dalam satu *pipeline* (Goertzel, keputusan DTMF, rekonstruksi *key*) serta menambah pemantauan simbol **payload** untuk mempermudah *debugging*.
- Memanfaatkan hasil **log** untuk iterasi perbaikan: *tuning* frekuensi, *valid pulse*, dan sinkronisasi *sampling*.

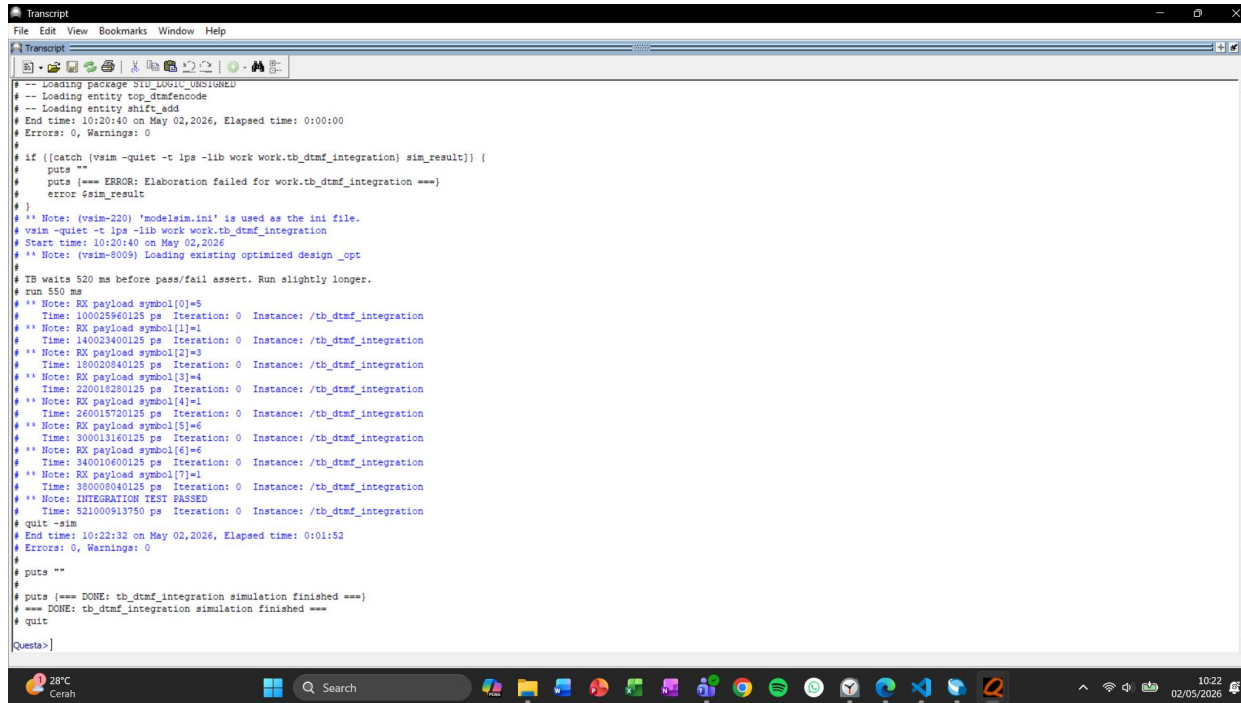
Simulasi *Testbench*



Hasil

Simulasi *Testbench* Berhasil

- Delapan segmen simbol *payload* kunci berhasil dideteksi dan direkonstruksi kembali tanpa pergeseran atau duplikasi data.
- Simulasi mencetak status "INTEGRATION TEST PASSED" dengan *0 errors* dan *0 warnings*.



```
Transcript
File Edit View Bookmarks Window Help

-- Loading package STD_LOGIC_UNRESOLVED
-- Loading entity top_dtmfencode
-- Loading entity shift_add
End time: 10:20:40 on May 02,2026, Elapsed time: 0:00:00
Errors: 0, Warnings: 0

if {[catch {vsim -quiet -t lps -lib work work.tb_dtmf_integration} sim_result]} {
    puts ""
    puts "==== ERROR: Elaboration failed for work.tb_dtmf_integration ==="
    error $sim_result
}

** Note: (vsim-220) 'modelsim.ini' is used as the ini file.
vsim -quiet -t lps -lib work work.tb_dtmf_integration
Start time: 10:20:40 on May 02,2026
** Note: (vsim-8009) Loading existing optimized design _opt

TB waits 520 ms before pass/fail assert. Run slightly longer.
run 550 ms
** Note: RX payload symbol[0]=5
Time: 100028960125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[1]=1
Time: 140023400125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[2]=3
Time: 180020840125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[3]=4
Time: 220018280125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[4]=1
Time: 260015720125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[5]=6
Time: 300013160125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[6]=6
Time: 340010600125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: RX payload symbol[7]=1
Time: 380008040125 ps Iteration: 0 Instance: /tb_dtmf_integration
** Note: INTEGRATION TEST PASSED
Time: 521000913750 ps Iteration: 0 Instance: /tb_dtmf_integration
quit -sim
End time: 10:22:32 on May 02,2026, Elapsed time: 0:01:52
Errors: 0, Warnings: 0

puts ""

puts "==== DONE: tb_dtmf_integration simulation finished ==="
==== DONE: tb_dtmf_integration simulation finished ====
quit

[Quarta>]
```


Kompilasi Desain Berhasil

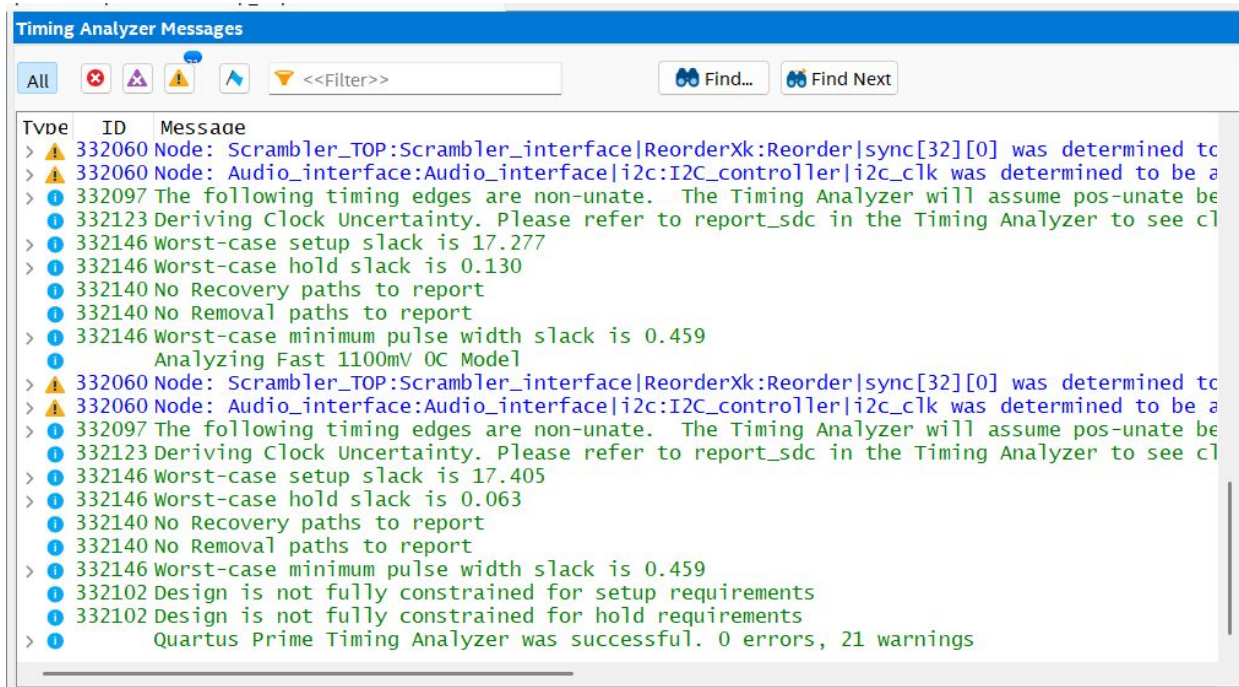
- Modul *Assembler* Quartus berhasil menerjemahkan desain ke dalam fail pemrograman perangkat (*bitstream*) untuk FPGA Altera Cyclone V (5CSXFC6D6F31C6) dengan status "Full Compilation was successful (0 errors, 192 warnings)".

The screenshot displays the Quartus Prime Lite Edition interface. The main window shows the 'Compilation Report - AcakCakap_Top' with a 'Flow Summary' table. The table indicates a successful compilation on May 02, 2026, for the Cyclone V device 5CSXFC6D6F31C6. The report lists various resources and their utilization, such as logic utilization (9,737 / 41,910), total pins (78 / 499), and total block memory bits (6,416 / 5,662,720). The bottom status bar shows '293000 Quartus Prime Full compilation was successful. 0 errors, 192 warnings'.

Flow Status	Successful - Sat May 02 10:07:13 2026
Flow Status	Successful - Sat May 02 10:07:13 2026
Quartus Prime Version	18.1.0 Build 625 09/12/2018 SJ Lite Edition
Revision Name	AcakCakap_Top
Top-level Entity Name	AcakCakap_Top
Family	Cyclone V
Device	5CSXFC6D6F31C6
Timing Models	Final
Logic utilization (in ALMs)	9,737 / 41,910 (23 %)
Total registers	11661
Total pins	78 / 499 (16 %)
Total virtual pins	0
Total block memory bits	6,416 / 5,662,720 (< 1 %)
Total DSP Blocks	61 / 112 (54 %)
Total HSSI RX PCSs	0 / 9 (0 %)
Total HSSI PMA RX Deserializers	0 / 9 (0 %)
Total HSSI TX PCSs	0 / 9 (0 %)
Total HSSI PMA TX Serializers	0 / 9 (0 %)
Total PLLs	1 / 15 (7 %)
Total DLLs	0 / 4 (0 %)

Analisis *Timing* Lolos

- Laporan TimeQuest Timing Analyzer menunjukkan tidak ada pelanggaran waktu (*setup* atau *hold violations*).
- Pada skenario ekstrem *Slow Model* (85°C), *worst-case setup slack* bernilai 15,744 ns dan *hold slack* 0,194 ns.

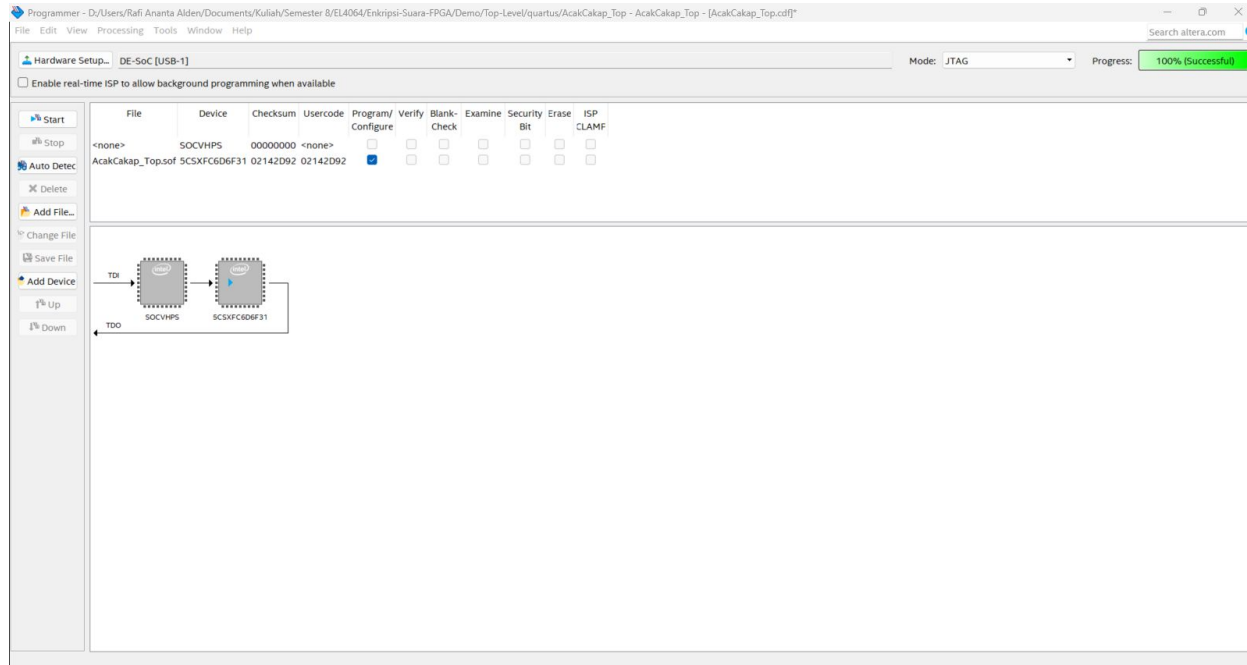


Timing Analyzer Messages

All [Icons] <<Filter>> Find... Find Next

Type	ID	Message
Warning	332060	Node: Scrambler_TOP:Scrambler_interface ReorderXk:Reorder sync[32][0] was determined to be non-unate.
Warning	332060	Node: Audio_interface:Audio_interface i2c:I2C_controller i2c_clk was determined to be non-unate.
Information	332097	The following timing edges are non-unate. The Timing Analyzer will assume pos-unate behavior.
Information	332123	Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainty values.
Information	332146	Worst-case setup slack is 17.277
Information	332146	Worst-case hold slack is 0.130
Information	332140	No Recovery paths to report
Information	332140	No Removal paths to report
Information	332146	Worst-case minimum pulse width slack is 0.459
Information		Analyzing Fast 1100mV OC Model
Warning	332060	Node: Scrambler_TOP:Scrambler_interface ReorderXk:Reorder sync[32][0] was determined to be non-unate.
Warning	332060	Node: Audio_interface:Audio_interface i2c:I2C_controller i2c_clk was determined to be non-unate.
Information	332097	The following timing edges are non-unate. The Timing Analyzer will assume pos-unate behavior.
Information	332123	Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer to see clock uncertainty values.
Information	332146	Worst-case setup slack is 17.405
Information	332146	Worst-case hold slack is 0.063
Information	332140	No Recovery paths to report
Information	332140	No Removal paths to report
Information	332146	Worst-case minimum pulse width slack is 0.459
Information	332102	Design is not fully constrained for setup requirements
Information	332102	Design is not fully constrained for hold requirements
Information		Quartus Prime Timing Analyzer was successful. 0 errors, 21 warnings

Pengunggahan *Bitstream* Berhasil



Pengujian *Hardware Loop Back Test* Berhasil

- Suara yang dimasukkan melalui mikrofon berhasil melewati seluruh *pipeline* perangkat keras (akuisisi analog-digital WM8731, algoritma FFT/IFFT di modul pengacak, serta algoritma Goertzel) dan berhasil direkonstruksi.
- Suara terdengar jernih dan dapat dipahami melalui *speaker*.



Temuan

Tuning Detektor Frekuensi tidak Tepat

- **Masalah:** Digit 7 dan 8 salah klasifikasi.
- **Penyebab:** Koefisien detektor ter-*tuning* ke bin frekuensi diskrit, bukan frekuensi DTMF standar.
- **Perbaikan:** Koefisien dikoreksi ke $2\cos(2\pi f/f_s)$ untuk setiap frekuensi target.
- **Hasil:** Semua 8 simbol *payload* terbaca benar.

Sinyal Valid Tidak Berbentuk Pulsa

- **Masalah:** Simbol terakumulasi ganda atau urutan tergeser.
- **Penyebab:** Sinyal valid bertahan beberapa siklus *clock*, bukan satu pulsa tunggal.
- **Perbaikan:** Sinyal valid dibatasi menjadi satu pulsa per kejadian deteksi.
- **Hasil:** Setiap simbol terekam tepat satu kali dengan urutan yang benar.

Domain Waktu Penerima Tidak Konsisten

- **Masalah:** Energi Goertzel tidak stabil antar simbol.
- **Penyebab:** Data masuk ke blok Goertzel tidak dibatasi laju 32 kHz.
- **Perbaikan:** Input receiver diikat ke penanda sampel 32 kHz.
- **Hasil:** Blok 640 sampel merepresentasikan tepat 20 ms, energi konsisten dan repeatable.

Desain dan Implementasi Sistem Pengamanan Komunikasi Suara Berbasis Enkripsi Frekuensi pada FPGA

Perbaikan *Frame Synchronization* di MATLAB
29 April - 6 Mei 2026

Tujuan

- Menemukan masalah *frame synchronization* di bagian MATLAB dan memperbaikinya
- Menentukan *golden reference frame synchronization* yang akan diimplementasikan ke *hardware*

Masalah dan Solusi MATLAB *Frame Synchronization* #1

Skema deteksi simbol “#” (*Flagging*) dan “3” (*Precision*) pada implementasi MATLAB tidak sama/sesuai dengan yang disampaikan di buku TA

1) Skema Deteksi “#”

- a) Jika nilai akumulasi *batch* sekarang 5 kali lebih besar daripada 24 batch sebelumnya, *count* akan bertambah 1 (Salah, seharusnya 3 kali lebih besar daripada 32 batch sebelumnya)
- b) Jika *count* mencapai 5 kali, sinyal *flag* deteksi salah satu frekuensi menyala (Benar)
- c) Jika semua sinyal *flag* frekuensi (*Flag* 941 Hz dan *flag* 1477 Hz) menyala, deteksi “#” berhasil (Benar)

2) Skema Deteksi “3”:

Menggunakan skema *global max value* dari nilai korelasi frekuensi 697 Hz, bukan membandingkan nilai korelasi sekarang terhadap nilai korelasi yang sebelumnya

Hasil Akhir:

Dilakukan pengubahan kondisi deteksi simbol-simbol tersebut sesuai yang ada di buku TA

Masalah dan Solusi MATLAB *Frame Synchronization* #2

Urutan simbol DTMF untuk *sync point* yang disimulasikan di MATLAB *frame synchronization* yang *floating point* masih menggunakan urutan yang tidak sesuai (“#33#”)

1. Akibat

- a. Pemilihan *batch* dan *frame* belum sesuai karena hasil simulasi dengan urutan simbol DTMF yang sesuai dengan *sync point* seharusnya itu memberikan hasil yang tidak sesuai harapan

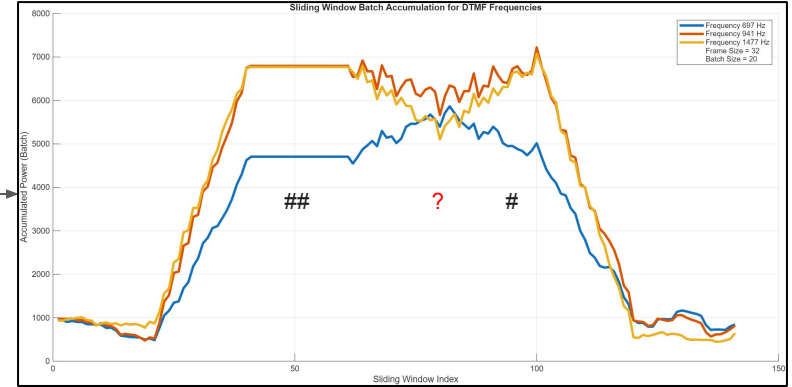
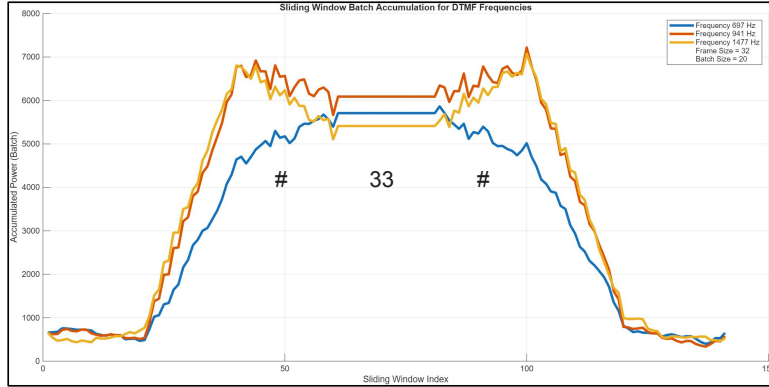
2. Solusi

- a. Mengganti urutan simbol DTMF ke urutan yang sesuai (“##3#”)
- b. Melakukan simulasi lagi dan menemukan kombinasi *batch-frame* yang memberikan hasil terbaik

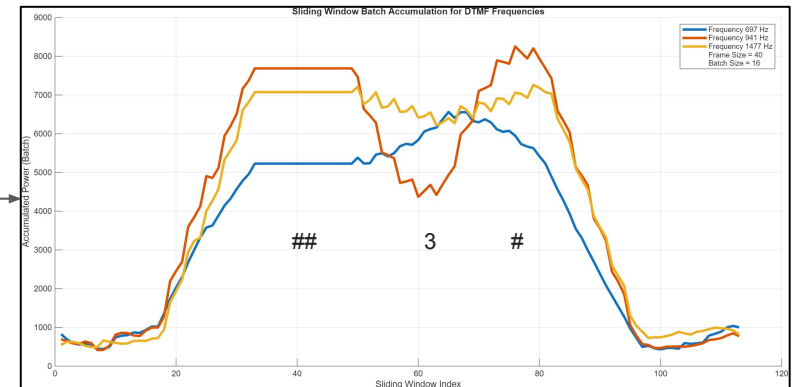
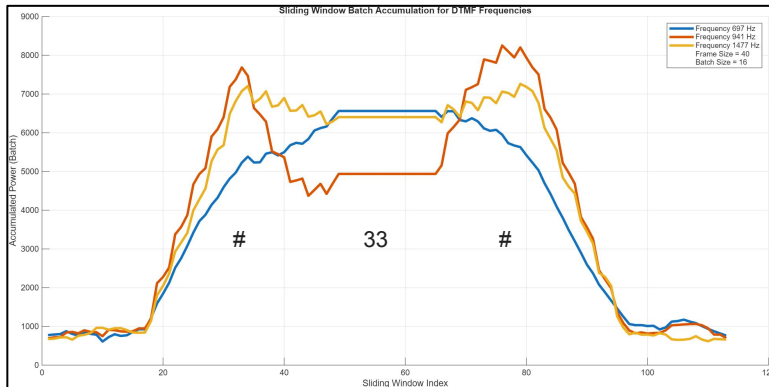
Hasil Akhir:

Setelah dilakukan iterasi simulasi, ditemukan kombinasi *frame-batch* yang sesuai, yaitu **40 sampel/frame** dan **16 frame/batch**, berdasarkan **hasil simulasi yang *floating-point***.

Masalah dan Solusi MATLAB *Frame Synchronization* #2



Dilakukan penyesuaian *frame-batch* agar deteksi “##3#” berhasil (32-20 -> 40-16).



Masalah dan Solusi MATLAB *Frame Synchronization* #3

Simulasi MATLAB yang *fixed-point* belum bisa-bisa karena nilai referensi LUT untuk sinyal sinusoidal dan sinyal cosinus dibuat untuk frekuensi 16kHz, bukan untuk frekuensi 32kHz

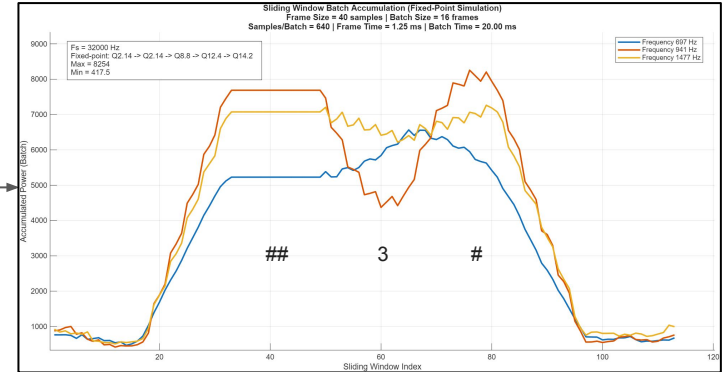
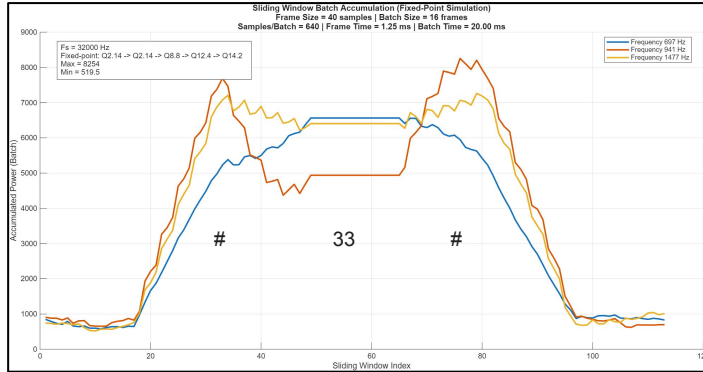
Hasil Akhir:

Referensi LUT untuk sinyal sinuoidal dan sinyal cosinus pada frekuensi 697Hz, 941Hz, dan 1477Hz diubah untuk frekuensi 32kHz dan simulasi MATLAB yang *fixed-point* dilakukan kembali untuk verifikasi hasil simulasi dengan hasil simulasi yang *floating-point*.

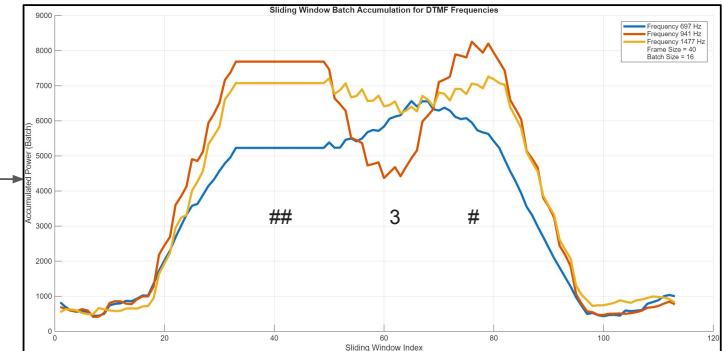
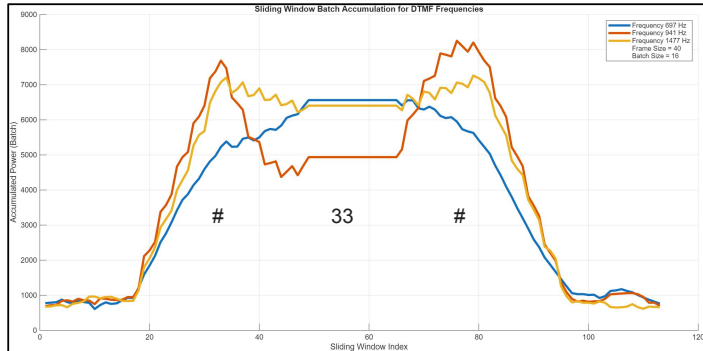
Catatan:

Penerapan solusi untuk mengatasi masalah-masalah MATLAB yang dijelaskan pada presentasi ini juga diterapkan ke semua kode MATLAB, baik yang *fixed-point* maupun yang *floating-point*.

Masalah dan Solusi MATLAB *Frame Synchronization* #3



Simulasi *Fixed-Point*



Simulasi *Floating-Point*

Masalah dan Solusi MATLAB *Frame Synchronization* #3

Nilai-nilai dari Simulasi *Floating Point*

```
[MULT SIN]
Range actual : [-1.989594, 1.999715]
Type        : Floating-point

[MULT COS]
Range actual : [-1.977258, 1.996441]
Type        : Floating-point

[ACC SIN]
Range actual : [-19.333923, 30.464978]
Type        : Floating-point

[ACC COS]
Range actual : [-18.392001, 22.261385]
Type        : Floating-point

[TOTAL POWER]
Range actual : [0.510337, 928.142885]
Type        : Floating-point

[BATCH SUM]
Range actual : [387.315428, 8252.797662]
Type        : Floating-point
```

Nilai-nilai dari Simulasi *Fixed-Point*

```
[MULT SIN]
Format      : Q2.14 (WL=16)
Range actual : [-1.98303, 1.99976]
Range theory : [-2.00000, 1.99994]
Saturation  : High=0 (0.000%), Low=0 (0.000%)

[MULT COS]
Format      : Q2.14 (WL=16)
Range actual : [-1.99078, 1.99646]
Range theory : [-2.00000, 1.99994]
Saturation  : High=0 (0.000%), Low=0 (0.000%)

[ACC SIN]
Format      : Q8.8 (WL=16)
Range actual : [-19.32812, 30.46484]
Range theory : [-128.00000, 127.99609]
Saturation  : High=0 (0.000%), Low=0 (0.000%)

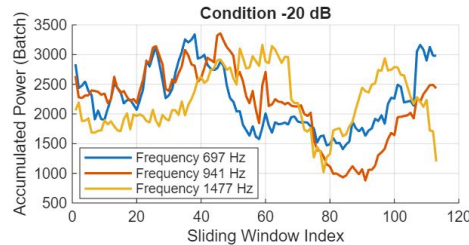
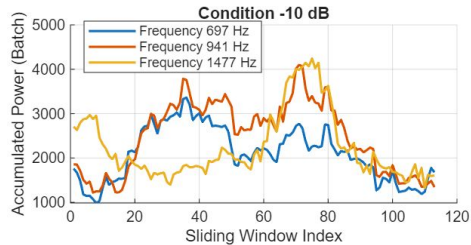
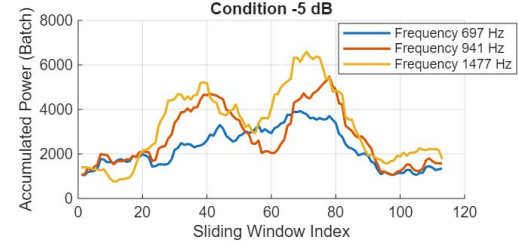
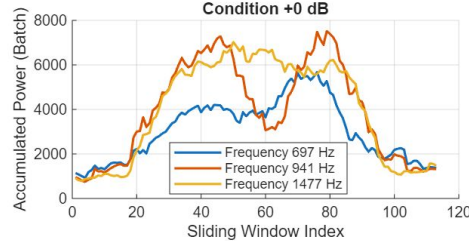
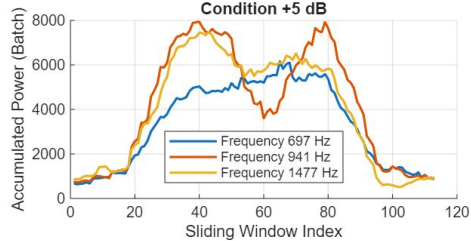
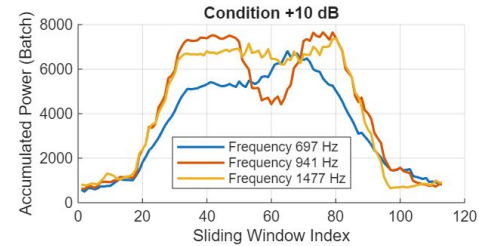
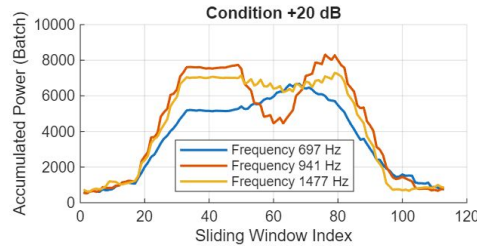
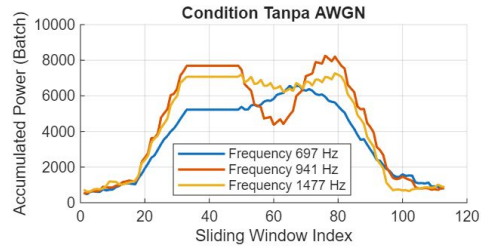
[ACC COS]
Format      : Q8.8 (WL=16)
Range actual : [-18.38672, 22.28125]
Range theory : [-128.00000, 127.99609]
Saturation  : High=0 (0.000%), Low=0 (0.000%)

[TOTAL POWER]
Format      : Q12.4 (WL=16)
Range actual : [0.06250, 928.12500]
Range theory : [-2048.00000, 2047.93750]
Saturation  : High=0 (0.000%), Low=0 (0.000%)

[BATCH SUM]
Format      : Q15.1 (WL=16)
Range actual : [417.50000, 8254.00000]
Range theory : [-16384.00000, 16383.50000]
Saturation  : High=0 (0.000%), Low=0 (0.000%)
```

Dengan fakta di atas, implementasi **hardware** dengan **fixed-point** tanpa **AWGN** akan menerapkan beberapa hal: **Q2.14** untuk **multiplier** dan **input sinyal**, **Q7.9** untuk **accumulator**, **Q12.4** untuk **power calculation**, dan **Q15.1** untuk **batch summation**.

Simulasi MATLAB Fixed-Point dengan AWGN #1



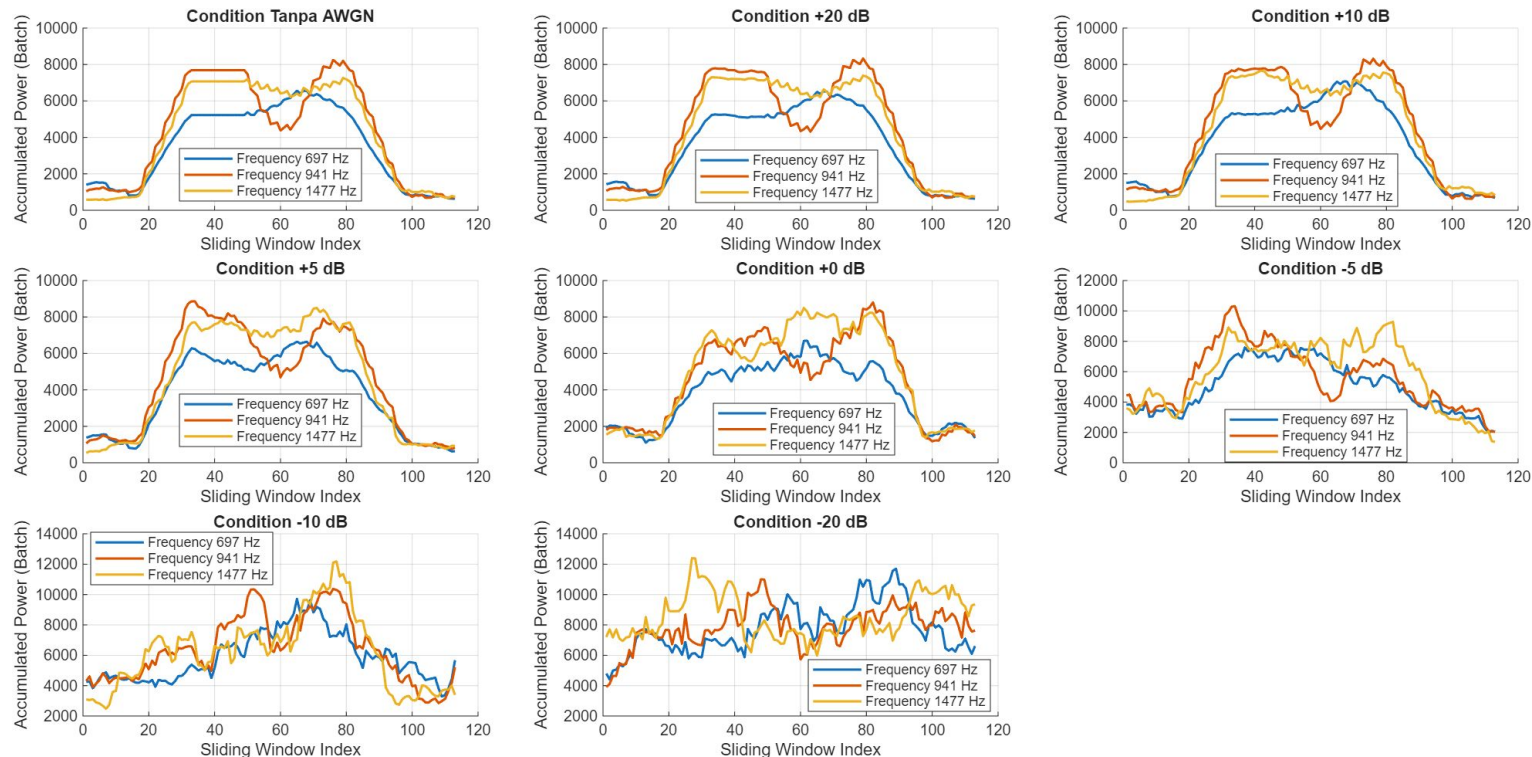
Hasil simulasi MATLAB dengan AWGN masih dapat sedikit terlihat sampai kondisi -5dB (Lebih bagus daripada yang disebutkan di buku TA, yaitu hanya sampai +5dB).

Simulasi MATLAB Fixed-Point dengan AWGN #1

[MULT SIN - +20 dB] Format : Q2.14 (WL=16) Range actual : [-1.99933, 1.99969] Range theory : [-2.00000, 1.99994] Saturation : High=0 (0.000%), Low=0 (0.000%)	[MULT SIN - +10 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99957] Range theory : [-2.00000, 1.99994] Saturation : High=0 (0.000%), Low=1 (0.007%)	[MULT SIN - +5 dB] Format : Q2.14 (WL=16) Range actual : [-1.99976, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=2 (0.013%), Low=0 (0.000%)	[MULT SIN - +0 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=6 (0.039%), Low=1 (0.007%)	[MULT SIN - -5 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=4 (0.026%), Low=1 (0.007%)
[MULT COS - +20 dB] Format : Q2.14 (WL=16) Range actual : [-1.99982, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=1 (0.007%), Low=0 (0.000%)	[MULT COS - +10 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=6 (0.039%), Low=4 (0.026%)	[MULT COS - +5 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=10 (0.065%), Low=3 (0.020%)	[MULT COS - +0 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=20 (0.130%), Low=9 (0.059%)	[MULT COS - -5 dB] Format : Q2.14 (WL=16) Range actual : [-2.00000, 1.99994] Range theory : [-2.00000, 1.99994] Saturation : High=41 (0.267%), Low=20 (0.130%)
[ACC SIN - +20 dB] Format : Q8.8 (WL=16) Range actual : [-19.44922, 31.00781] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC SIN - +10 dB] Format : Q8.8 (WL=16) Range actual : [-20.62109, 31.08203] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC SIN - +5 dB] Format : Q8.8 (WL=16) Range actual : [-19.79297, 33.00391] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC SIN - +0 dB] Format : Q8.8 (WL=16) Range actual : [-21.58203, 31.85938] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC SIN - -5 dB] Format : Q8.8 (WL=16) Range actual : [-18.33203, 30.80859] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)
[ACC COS - +20 dB] Format : Q8.8 (WL=16) Range actual : [-18.89453, 22.44922] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC COS - +10 dB] Format : Q8.8 (WL=16) Range actual : [-20.12891, 23.24219] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC COS - +5 dB] Format : Q8.8 (WL=16) Range actual : [-18.05078, 24.55078] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC COS - +0 dB] Format : Q8.8 (WL=16) Range actual : [-24.03516, 21.35547] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)	[ACC COS - -5 dB] Format : Q8.8 (WL=16) Range actual : [-21.62891, 24.62109] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)
[TOTAL POWER - +20 dB] Format : Q12.4 (WL=16) Range actual : [0.00000, 961.50000] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)	[TOTAL POWER - +10 dB] Format : Q12.4 (WL=16) Range actual : [1.00000, 968.25000] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)	[TOTAL POWER - +5 dB] Format : Q12.4 (WL=16) Range actual : [0.25000, 1108.56250] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)	[TOTAL POWER - +0 dB] Format : Q12.4 (WL=16) Range actual : [0.62500, 1024.56250] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)	[TOTAL POWER - -5 dB] Format : Q12.4 (WL=16) Range actual : [0.06250, 1012.43750] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)
[BATCH SUM - +20 dB] Format : Q15.1 (WL=16) Range actual : [516.00000, 8310.50000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)	[BATCH SUM - +10 dB] Format : Q15.1 (WL=16) Range actual : [497.00000, 7649.00000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)	[BATCH SUM - +5 dB] Format : Q15.1 (WL=16) Range actual : [509.00000, 7937.00000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)	[BATCH SUM - +0 dB] Format : Q15.1 (WL=16) Range actual : [747.50000, 7524.00000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)	[BATCH SUM - -5 dB] Format : Q15.1 (WL=16) Range actual : [762.00000, 6591.00000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)

Ada saturasi pada hasil *multiplier* sehingga perlu diatur lagi konfigurasi *fixed-point* pada *multiplier* dan *input* sinyal (Q2.14 -> Q3.13).

Simulasi MATLAB Fixed-Point dengan AWGN #2



Dengan *input* sinyal dan *multiplier* yang Q3.13, hasil simulasi MATLAB dengan AWGN kurang lebih sama saja dengan *input* sinyal dan *multiplier* yang Q2.14.

Simulasi MATLAB Fixed-Point dengan AWGN #2

<pre>[MULT SIN - +20 dB] Format : Q3.13 (WL=16) Range actual : [-2.19849, 2.22144] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT SIN - +10 dB] Format : Q3.13 (WL=16) Range actual : [-2.63098, 2.89697] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT SIN - +5 dB] Format : Q3.13 (WL=16) Range actual : [-3.67627, 3.85559] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT SIN - +0 dB] Format : Q3.13 (WL=16) Range actual : [-3.97900, 3.99805] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT SIN - -5 dB] Format : Q3.13 (WL=16) Range actual : [-3.99988, 3.99988] Range theory : [-4.00000, 3.99988] Saturation : High=3 (0.020%), Low=0 (0.000%)</pre>
<pre>[MULT COS - +20 dB] Format : Q3.13 (WL=16) Range actual : [-2.20569, 2.21851] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT COS - +10 dB] Format : Q3.13 (WL=16) Range actual : [-3.02722, 2.85608] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT COS - +5 dB] Format : Q3.13 (WL=16) Range actual : [-3.77087, 3.53076] Range theory : [-4.00000, 3.99988] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[MULT COS - +0 dB] Format : Q3.13 (WL=16) Range actual : [-4.00000, 3.99988] Range theory : [-4.00000, 3.99988] Saturation : High=2 (0.013%), Low=3 (0.020%)</pre>	<pre>[MULT COS - -5 dB] Format : Q3.13 (WL=16) Range actual : [-4.00000, 3.99988] Range theory : [-4.00000, 3.99988] Saturation : High=12 (0.078%), Low=8 (0.052%)</pre>
<pre>[ACC SIN - +20 dB] Format : Q8.8 (WL=16) Range actual : [-19.73828, 31.20312] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC SIN - +10 dB] Format : Q8.8 (WL=16) Range actual : [-20.80078, 32.13281] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC SIN - +5 dB] Format : Q8.8 (WL=16) Range actual : [-20.98438, 33.20312] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC SIN - +0 dB] Format : Q8.8 (WL=16) Range actual : [-23.87500, 38.38672] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC SIN - -5 dB] Format : Q8.8 (WL=16) Range actual : [-27.71094, 44.19531] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>
<pre>[ACC COS - +20 dB] Format : Q8.8 (WL=16) Range actual : [-19.14062, 22.35156] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC COS - +10 dB] Format : Q8.8 (WL=16) Range actual : [-20.40625, 21.68359] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC COS - +5 dB] Format : Q8.8 (WL=16) Range actual : [-22.24609, 23.66016] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC COS - +0 dB] Format : Q8.8 (WL=16) Range actual : [-24.00781, 23.30469] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[ACC COS - -5 dB] Format : Q8.8 (WL=16) Range actual : [-29.44531, 30.65625] Range theory : [-128.00000, 127.99609] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>
<pre>[TOTAL POWER - +20 dB] Format : Q12.4 (WL=16) Range actual : [0.12500, 974.06250] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[TOTAL POWER - +10 dB] Format : Q12.4 (WL=16) Range actual : [0.87500, 1033.68750] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[TOTAL POWER - +5 dB] Format : Q12.4 (WL=16) Range actual : [0.87500, 1102.56250] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[TOTAL POWER - +0 dB] Format : Q12.4 (WL=16) Range actual : [0.93750, 1473.81250] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[TOTAL POWER - -5 dB] Format : Q12.4 (WL=16) Range actual : [0.93750, 1963.68750] Range theory : [-2048.00000, 2047.93750] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>
<pre>[BATCH SUM - +20 dB] Format : Q15.1 (WL=16) Range actual : [528.50000, 8332.50000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[BATCH SUM - +10 dB] Format : Q15.1 (WL=16) Range actual : [466.00000, 8318.00000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[BATCH SUM - +5 dB] Format : Q15.1 (WL=16) Range actual : [538.50000, 8860.50000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[BATCH SUM - +0 dB] Format : Q15.1 (WL=16) Range actual : [1100.50000, 8794.50000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>	<pre>[BATCH SUM - -5 dB] Format : Q15.1 (WL=16) Range actual : [1368.00000, 10310.50000] Range theory : [-16384.00000, 16383.50000] Saturation : High=0 (0.000%), Low=0 (0.000%)</pre>

Saturasi terjadi pada kondisi SNR +0dB. Jika ingin dioptimalkan lebih lanjut, dapat gunakan konfigurasi *fixed-point* dengan *word length* atau *bit size* yang lebih besar daripada 16-bit.

Dengan demikian, akan diterapkan konfigurasi *fixed-point* di *hardware* dengan deskripsi: **Q3.13** untuk *multiplier* dan *input* sinyal, **Q7.9** untuk *accumulator*, **Q12.4** untuk *power calculation*, dan **Q15.1** untuk *batch summation*.

Rencana

- Mengimplementasikan *hardware* dari *frame synchronization*
- Mengintegrasikan *frame synchronization* ke *top-level* DTMF satu arah
- Melakukan pengujian *hardware* dari *top-level* DTMF satu arah yang sudah memiliki *frame synchronization* terbaru (Jika mungkin)